



Embedded Systems



CORTEX-M4 INSTRUCTIONS

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SHIFT OPERATIONS

Register shift operations move the bits in a register left or right by a specified number of bits, the shift length. Register shift can be performed directly by the instructions ASR, LSR, LSL, and ROR and the result is written to a destination register.

01 ASR Instruction

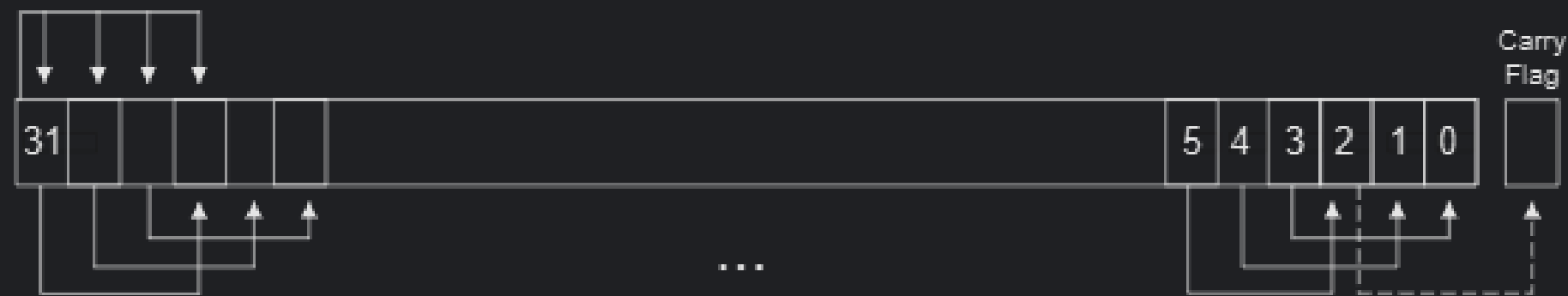
Arithmetic shift right by n bits moves the left-hand $32-n$ bits of the register R_m , to the right by n places, into the right-hand $32-n$ bits of the result, and it copies the original bit[31] of the register into the left-hand n bits of the result.

You can use the ASR operation to divide the signed value in the register R_m by 2^n , with the result being rounded towards negative-infinity.

When the instruction is ASRS the carry flag is updated to the last bit shifted out, bit[$n-1$], of the register R_m .

ASR & ASRS

Figure 3.1. ASR #3



SIMPLE FIGURE OF ASRS



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Thank's For Listening

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